

EE201: Circuit Theory I

Syllabus — Fall 2026–2027

Mert Ankaralı
Section 2

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Section 2 material: github.com/mertankarali/Lecture-Notes

Instructors — EE201, Fall 2026–2027

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You are in **Section 2** — office A-410/D-104, mertan@metu.edu.tr

Teaching assistants

EE201 — theory

Anıl Gülbeden

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EE201/EE213 — laboratory

Alp Gürsoy Eda Özkaynar Orhun Koç Civan Serhat Çevik
Elif Çağla Tay Enfal Çabuk Adem Utku Atasayar
Özgür Gülsuna Alper Alka

Laboratory questions go to the laboratory assistants first; theory questions to Anıl or to me.

Course material — common to all sections

1. **Alexander & Sadiku**, *Fundamentals of Electric Circuits*, McGraw-Hill, 2016.
main textbook
2. **Chua, Desoer & Kuh**, *Linear and Nonlinear Circuits*, McGraw-Hill, 1987.
reference
3. Lecture notes and videos by **Emre Tuna** — users.metu.edu.tr/etuna/ee201

ODTÜCLASS —

The official channel for *all* sections: announcements, the detailed course outline, grades, exam and laboratory information, and the common material above.

Reading the textbook *before* the lecture is worth more than reading it after.

And, for Section 2: a course repository

`github.com/mertankarali/Lecture-Notes`

folder: METU-EE201

- My lecture notes, in-class examples, extra problems with solutions, code files.
- **Public** — no account needed to read or download.
- Updated **during** the semester, usually before each lecture.
- Clone it and `git pull`, or just re-download — but do not work from a copy you downloaded in October.
- Found a typo or an error? Open an issue or send me an e-mail.

What is where

ODTÜCLASS (all sections)

- official announcements
- detailed course outline
- grades
- exam and laboratory info
- common notes and videos

Repository (Section 2)

- my lecture notes
- in-class examples
- extra problems + solutions
- code / simulation files

Announcements are *always* on ODTÜCLASS. The repository is for content.

Grading

Component	Weight
Attendance	5 %
Midterm Exam 1	20 %
Midterm Exam 2	20 %
Final Exam	35 %
Laboratory	20 %
Total	100 %

The laboratory is one fifth of your grade — and a gate, see the next slide.

Attendance in Section 2

- Taken with a **QR code on ODTÜCLASS** — bring a device, stay logged in.
- **Once per class meeting** — once in a one-hour meeting, once in a two-hour meeting. Not once per lecture hour.
- Normally **at the beginning**. In a two-hour meeting, if I give a break, I may open it **at the break** instead.
- I sometimes teach two hours as one block, sometimes finish early, and I do not always open it at the break.

So

There is no pattern to rely on. Skipping the first hour and arriving at the break is **your own risk** — if attendance was taken while you were not in the room, that meeting is lost and it is not repeated.

Final examination policy — how to get an NA

You will **not be admitted to the final exam** and will receive **NA** if:

1. your attendance is below **40 %**;
2. you miss **any midterm** without a valid excuse;
3. you miss **any laboratory experiment** without a valid excuse;
4. you miss the **Laboratory Final Exam** without a valid excuse.

Read this as

Show up, and if something happens, report it *with a valid excuse and on time*.

Make-up policy

- A make-up compensates for **one** missing exam or laboratory.
- Missing more than one $\Rightarrow$ multiple make-ups.
- Make-ups may be scheduled **before or after** the final exam.
- Dates are announced on **ODTÜCLASS**; the laboratory make-up is announced **separately**.

Follow the ODTÜCLASS announcements — "I did not know" is not an excuse.

If you have taken EE213 or the old EE201

- **Took EE213, have a grade, now registered to the new EE201:**
your grade comes from the EE201 *theoretical* part (100 %); the previous laboratory grade is **not** included.
- **Failed EE213 and take only the EE213 laboratory:**
your grade comes from the laboratory work only, as before.
- **New EE201 (theory + laboratory)** is opened only for students taking the course for the **first time**.

Unsure which case you are in? Talk to me after the lecture.

Course outline — the whole semester

Topic	Hours
1. Basic Concepts	6
2. Linear Time-Invariant Resistive Circuits	10
3. Nonlinear Resistive Circuits	3
4. Operational Amplifier Circuits	7
5. Dynamic Elements	5
6. First Order Circuits	11
Total	42

Next three slides: what is inside each of these.

Outline (1/3) — resistive circuits

1. Basic Concepts

6 hrs

Introduction; lumped elements and lumped circuits; interconnection equations; branch relations; basic lumped elements; circuit analysis.

2. Linear Time-Invariant Resistive Circuits

10 hrs

LTI resistive elements: resistors, dependent sources, ideal transformers; analysis methods; one-port circuits; two-ports.

This is where the language of the course is built — do not fall behind here.

Outline (2/3) — nonlinearity and op-amps

3. Nonlinear Resistive Circuits

3 hrs

Nonlinear resistive elements and circuits; load-line analysis; piecewise-linear resistive circuits.

4. Operational Amplifier Circuits

7 hrs

Operational amplifiers; basic op-amp circuits; more realistic op-amp models; miscellaneous resistive op-amp circuits; op-amps with piecewise-linear elements.

Outline (3/3) — dynamics

5. Dynamic Elements

5 hrs

Ramp, step and impulse functions; capacitors; inductors.

6. First Order Circuits

11 hrs

First order linear ODEs with constant coefficients; simple LTI RC circuit; simple LTI RL circuit; miscellaneous LTI first order circuits; piecewise-linear first order circuits.

From algebraic equations to differential equations — the heart of the course.

Laboratory outline

Exp.	Title
#1	Introduction to Voltage, Current and Resistance Measurements
#2	Signal Generators and Oscilloscopes
#3	LTspice Workshop (<i>computer experiment</i>)
#4	Resistive Circuits
#5	Operational Amplifiers
#6	First Order Circuits

The experiments follow the theory — and missing one without an excuse means NA.

Before the next lecture

1. Check that you are enrolled on **ODTÜCLASS** for EE201 — and for EE213 as well, if you take the laboratory separately.
2. Open the repository and bookmark it:
`github.com/mertankarali/Lecture-Notes`
3. Get the textbook (Alexander & Sadiku) and skim Chapter 1.
4. Note the detailed outline on ODTÜCLASS.

Questions?

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